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In [5]: import pandas as pd
df=pd.read_csv('breast_cancer_clinical3.csv')
print('breast_cancer_clinical3.csv读取成功')
print('-----')
import statsmodels
import statsmodels.api as sm
import numpy as np
variables=['Age','Tumor_Size','Ki67','ER','PR','HER2']
uni_results=[]
for variable in variables:
    y=df['Lymph_Node']
    X=df[[variable]]
    X=sm.add_constant(X)
    model=sm.Logit(y,X).fit(dis=False)
    OR=np.exp(model.params[variable])
    CI=np.exp(model.conf_int().loc[variable])
    p=model.pvalues[variable]
    uni_results.append([
        variable,
        OR,
        CI[0],
        CI[1],
        p
    ])
uni_table=pd.DataFrame(
    uni_results,
    columns=[
        'Variable',
        'Uni_OR',
        'Uni_CI_low',
        'Uni_CI_high',
        'Uni_P'
    ])
print('Uni_table:')
print(uni_table)
print('-----')
multi_results=[]
y=df['Lymph_Node']
X=df[['Age','Tumor_Size','Ki67','ER','PR','HER2']]
X=sm.add_constant(X)
model=sm.Logit(y,X).fit(dis=False)
for variable in variables:
    OR=np.exp(model.params[variable])
    CI=np.exp(model.conf_int().loc[variable])
    p=model.pvalues[variable]
    multi_results.append([
        variable,
        OR,
        CI[0],
        CI[1],
        p
    ])
multi_table=pd.DataFrame(
    multi_results,
    columns=[
        'Variable',
        'Multi_OR',
        'Multi_CI_low',

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        'Multi_CI_high',
        'Multi_P'
    ])
print('multi_table:')
print(multi_table)
print('-----')
final_table=pd.merge(uni_table,multi_table,on='Variable')
print('Final_table:')
print(final_table)
print('-----')
final_table['Uni_OR_CI']=(
    final_table['Uni_OR'].round(2).astype(str)
    +
    '('
    +
    final_table['Uni_CI_low'].round(2).astype(str)
    +
    '_'
    +
    final_table['Uni_CI_high'].round(2).astype(str)
    +
    ')'
)
final_table['Multi_OR_CI']=(
    final_table['Multi_OR'].round(2).astype(str)
    +
    '('
    +
    final_table['Multi_CI_low'].round(2).astype(str)
    +
    '_'
    +
    final_table['Multi_CI_high'].round(2).astype(str)
    +
    ')'
)
paper_table=final_table[['Variable','Uni_OR_CI','Uni_P','Multi_OR_CI','Multi_P']]
print('paper_table:')
print(paper_table)
print('-----')
paper_table.to_excel('paper_table.xlsx')
print('paper_table.xlsx保存成功')
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breast_cancer_clinical3.csv读取成功

Uni_table:

	Variable	Uni_OR	Uni_CI_low	Uni_CI_high	Uni_P
0	Age	1.329203	1.036229	1.705010	0.025084
1	Tumor_Size	13.518047	1.534596	119.078654	0.018989
2	Ki67	1.010267	0.980639	1.040791	0.501194
3	ER	0.104167	0.009509	1.141126	0.064044
4	PR	0.063492	0.007072	0.570034	0.013821
5	HER2	9.333333	1.193455	72.990686	0.033296

multi_table:

	Variable	Multi_OR	Multi_CI_low	Multi_CI_high	Multi_P
0	Age	4.406498e-01	4.137368e-02	4.693135e+00	0.497150
1	Tumor_Size	1.515478e+06	7.026552e-05	3.268563e+16	0.241103
2	Ki67	1.061867e+00	9.717194e-01	1.160378e+00	0.184782
3	ER	3.605624e+04	6.178724e-05	2.104079e+13	0.308264
4	PR	2.169745e-02	1.618976e-06	2.907884e+02	0.429511
5	HER2	8.366226e-01	6.838726e-07	1.023491e+06	0.980101

Final_table:

	Variable	Uni_OR	Uni_CI_low	Uni_CI_high	Uni_P	Multi_OR	\
0	Age	1.329203	1.036229	1.705010	0.025084	4.406498e-01	
1	Tumor_Size	13.518047	1.534596	119.078654	0.018989	1.515478e+06	
2	Ki67	1.010267	0.980639	1.040791	0.501194	1.061867e+00	
3	ER	0.104167	0.009509	1.141126	0.064044	3.605624e+04	
4	PR	0.063492	0.007072	0.570034	0.013821	2.169745e-02	
5	HER2	9.333333	1.193455	72.990686	0.033296	8.366226e-01	

	Multi_CI_low	Multi_CI_high	Multi_P
0	4.137368e-02	4.693135e+00	0.497150
1	7.026552e-05	3.268563e+16	0.241103
2	9.717194e-01	1.160378e+00	0.184782
3	6.178724e-05	2.104079e+13	0.308264
4	1.618976e-06	2.907884e+02	0.429511
5	6.838726e-07	1.023491e+06	0.980101

paper_table:

	Variable	Uni_OR_CI	Uni_P	\
0	Age	1.33(1.04-1.71)	0.025084	
1	Tumor_Size	13.52(1.53-119.08)	0.018989	
2	Ki67	1.01(0.98-1.04)	0.501194	
3	ER	0.1(0.01-1.14)	0.064044	
4	PR	0.06(0.01-0.57)	0.013821	
5	HER2	9.33(1.19-72.99)	0.033296	

	Multi_OR_CI	Multi_P
0	0.44(0.04-4.69)	0.497150
1	1515477.65(0.0-3.2685627132824944e+16)	0.241103
2	1.06(0.97-1.16)	0.184782
3	36056.24(0.0-21040793759940.05)	0.308264
4	0.02(0.0-290.79)	0.429511
5	0.84(0.0-1023490.85)	0.980101

paper_table.xlsx保存成功

In []: